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Synthetic Methods

Enantioselective Allenation of Terminal Alkynes Catalyzed by Copper Halides of Mixed Oxidation States and Its Application to the Total Synthesis of Scorodonin

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Dedicated to Professor E. Peter Kündig on the occasion of his 75th birthday

Abstract: *Naturally occurring conjugated allenynes are of general interest owing to their potent and various biological activities. The 1,5-H transfer of alka-1,4-diyn-3-yl amines would be one of the most straightforward yet challenging approaches to such compounds since, in principle, two regioisomers may be formed involving two C−C triple bonds. Herein, a catalytic recipe of copper halides with mixed oxidation states, i.e., CuCl/CuBr$_2$, has been identified to address the issues of the side reaction of conjugate addition and the selectivity of 1,5-H transfer of alka-1,4-diyn-3-yl amines in EATA (enantioselective allenation of terminal alkynes) reaction involving 2-alkynals. This method provided various allenynes with excellent enantioselectivities, and was also applied to the first highly enantioselective total synthesis of natural product scorodonin. Mechanistic studies and DFT calculations elucidated the regioselectivity for the observed 1,5-H transfer.*

Introduction

Allenes exist in over 150 natural products[1] and are a very important class of basic chemicals for organic synthesis,[2] pharmaceuticals,[1b] and materials science.[3] Of particular interest, the conjugated allenynes are one of the most conspicuous structural unit in these natural products (Figure 1 A)[1b,4] and highlight the potential utility in organic synthesis.[5] However, the lack of an efficient and flexible synthesis has prevented its further synthetic applications and biological studies. Much efforts have been devoted over the past few decades to develop new methods for the efficient synthesis of allenes.[6,7] In particular, the allenation of terminal alkynes (ATA) reaction promoted with transition metals, such as Cu,[8] Zn,[9] Au,[10] has received great attention (Figure 1 B). However, such reactions with 2-alkynals are still problematic and have never been reported since there are several challenges to be solved: (1) the potential conjugate addition of amines with 2-alkynals; (2) the site-selectivity of 1,5-H transfer[11,12] within the 1,4-alkadiyn-3-yl amines (**Int**) bearing two different C−C triple bonds; (3) the suitable catalyst to initiate the reaction with good efficiency and enantiocontrol. Here we have identified a catalytic recipe of copper halides with mixed oxidation states to address such selectivity issues on the EATA reaction of 2-alkynals and terminal alkynes in the presence of the commercially available (*S*)- or (*R*)-α,α-diphenylprolinol (Figure 1 C). Remarkably, this new recipe is highlighted by the first highly enantioselective concise total synthesis of both enantiomers of scorodonin.

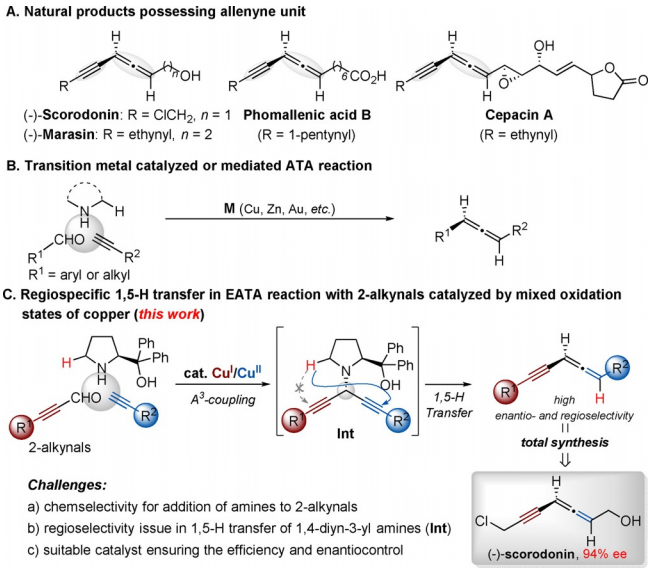


Figure 1. ATA reaction and the possible enantioselective synthesis of allenynes through regiospecific 1,5-H transfer.

Results and Discussion

After some trial and error, we turned our attention to the reaction of 3-*t*-butyldimethylsilylpropynal **1a** and tertiary propargylic alcohol **2a** in the presence of (*S*)-α,α-diphenylprolinol ((*S*)-**3**) in 1,4-dioxane at 80 °C. An extensive screen of

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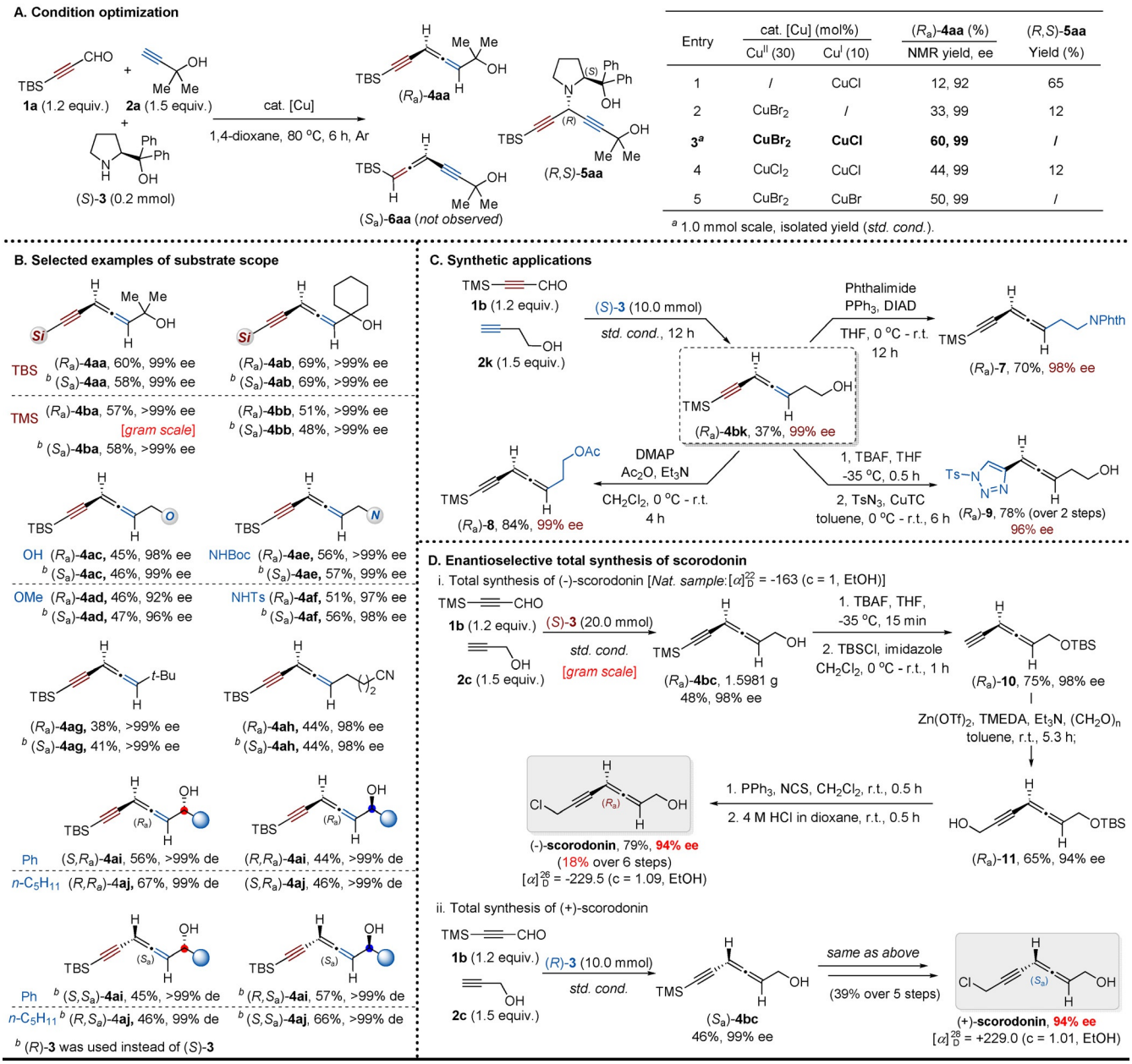


Figure 2. Development of CuCl/$CuBr_2$ co-catalyzed EATA of 2-alkynals. TBS = *t*-butyldimethylsilyl; TMS = trimethylsilyl; Boc = *t*-butoxycarbonyl; Ts = *p*-toluenesulfonyl; DIAD = diisopropyl azodicarboxylate; THF = tetrahydrofuran; Phth = phthaloyl; DMAP = *N*,*N*-4-dimethylaminopyridine; TBAF = tetra-*n*-butylammonium fluoride; CuTC = copper(I)-thiophene-2-carboxylate; TMEDA = *N*,*N*,*N*′,*N*′-tetramethylethylenediamine; NCS = *N*-chlorosuccinimide.

metal catalysts demonstrated that the copper halides of mixed oxidation states, i.e., CuCl/$CuBr_2$ (10/30 mol%), were optimal catalyst for this EATA reaction to afford the desired allenyne (R_a)-**4aa** in 60% yield with 99% *ee*, while the regioisomeric allene product (S_a)-**6aa** was not observed (Figure 2A). Mono-metallic or other mixed couples showed low efficiency and gave the allenyne (R_a)-**4aa** together with 1,4-diyn-3-yl amine intermediate (*R*,*S*)-**5aa** (see Table S1 in Supporting Information).

With the optimal reaction conditions identified, we next studied the scopes of the EATA reaction with 3-silylpropynals **1a** and **1b**. A wide range of terminal alkynes containing diverse functional groups and alkyl substituents were tolerated, delivering the corresponding allenynes with high enantioselectivities (up to >99% *ee*). By applying chiral secondary propargylic alcohols (*S*)-**2i** along with (*R*)- or (*S*)-**3**, (S,R_a)-**4ai** and (S,S_a)-**4ai** were produced in high diastereoselectivity. Likewise, the other two stereoisomers (R,R_a)- and (R,S_a)-**4ai** could be also obtained through a similar strategy by starting from (*R*)-**2i**. Secondary alkyl substituted propargylic alcohols **2j** could also be employed, affording all the diastereomers of **4aj** in moderate yield with high diastereoselectivity (up to >99% de) (Figure 2B, for the full scopes, see Table S2 in Supporting Information).

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Synthetic applications of these chiral allenynes were demonstrated as shown in Figure 2C: the reaction may be easily scaled up to 10 mmol, affording allenyne (R_a)-**4bk** with 99% *ee*. The Mitsunobu reaction of (R_a)-**4bk** with phthalimide proceeded smoothly to afford (R_a)-**7** in 70% yield with 98% *ee*. This primary allenol could also be easily transformed to the corresponding ester (R_a)-**8** without racemization. In addition, the triazole-tethered allenes (R_a)-**9** could be successfully assembled in high *ee* via desilylation followed by cycloaddition with azide (see details in Supporting Information).[13]

Scorodonin, a biologically active metabolite isolated from submerged cultures of the mushroom *Marasmius scorodonius*, showed interesting antibacterial and antifungal activities.[4d] Due to the presence of the highly sensitive conjugate allenyne unit, its highly enantioselective synthesis is challenging and has not been realized.[14] We reasoned that the EATA reaction of 2-alkynals would be a straightforward approach to the projected compound. As shown in Figure 2D-i, our synthesis commenced with the EATA reaction of 2-alkynal **1b** with propargyl alcohol **2c**, which easily furnished allenyne (R_a)-**4bc** on a gram scale with 98% *ee*. Subsequent desilylation and TBS protection delivered enantioenriched (R_a)-**10** in 75% yield over two steps. Next, the hydroxymethylation of terminal allenyne (R_a)-**10** proved to be unexpectedly challenging, with the commonly used base/paraformaldehyde conditions failing to realize this transformation (see details in Supporting Information). Ultimately, a modified alkynylation of (R_a)-**10** with paraformaldehyde in the presence of Zn(OTf)$_2$/TMEDA/Et$_3$N successfully generated (R_a)-**11** in 65% yield with 94% *ee*.[15] Finally, an Appel chlorination[16] of the hydroxyl group with NCS, followed by acid-mediated desilylation afforded (−)-scorodonin with 94% *ee*. Likewise, the synthesis of (+)-scorodonin was executed following the same synthetic route just by replacing (*S*)-**3** with (*R*)-**3** in the step of EATA reaction with a similar yield and enantioselectivity (Figure 2D-ii, see details in Supporting Information).

To gain insight into the reaction mechanism, a series of control experiments were carried out (Figure 3A): When alka-1,4-diyn-3-yl amine (*R*,*S*)-**5ba** was treated with the combination of a catalytic amount each of $CuBr_2$ and CuCl, allenyne (R_a)-**4ba** could be obtained in 65% yield with >99% *ee*, confirming that (*R*,*S*)-**5ba** was the key intermediate. Interestingly, the use of $CuBr_2$ alone afforded the product (R_a)-**4ba** in comparable yield and same *ee*; when 10 mol% CuCl was added, (R_a)-**4ba** was afforded in 14% yield with

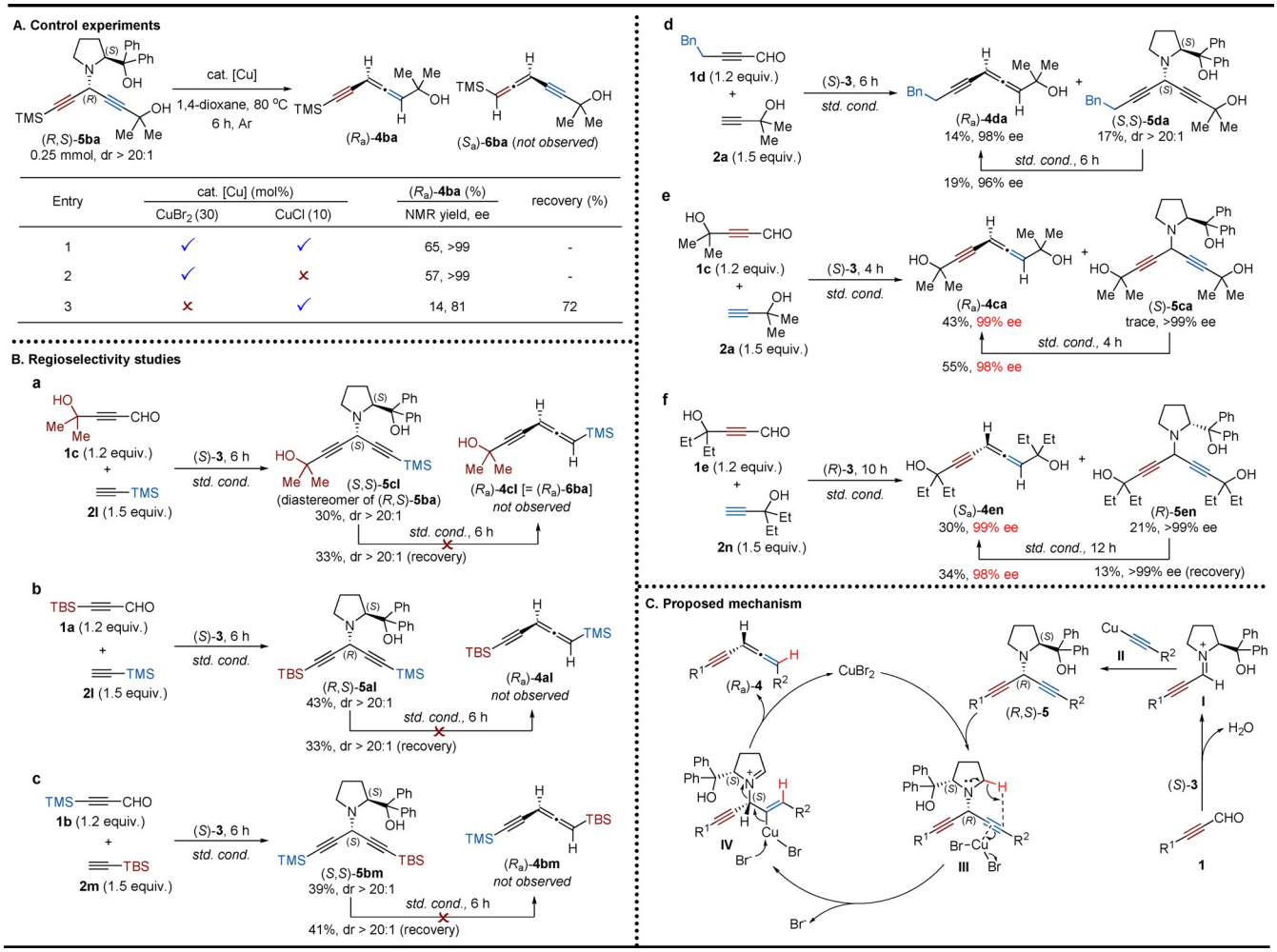


Figure 3. Preliminary mechanistic studies.

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81 % *ee* together with 72 % of recovered (*R*,*S*)-**5ba**. Based on these studies and the results demonstrated in entries 1 and 2 of Figure 2A, we concluded that the first step was realized with the combination of $CuBr_2$ and CuCl, while $CuBr_2$ worked efficiently for 1,5-H transfer ensuring a high *ee* for the final product by overwriting the role of CuCl. It should be noted that the regioisomer (S_a)-**6ba** was never observed in all of the above transformations. This indicated the 1,5-H transfer reaction is regiospecific to the C−C triple bond from the terminal alkynes.

In sharp contrast with the exclusive formation of (R_a)-**4ba** from **1b** and **2a**, when the substituents in both the terminal alkyne and 2-alkynal were switched, i.e., applying 4-hydroxy-4-methyl-2-pentynal **1c** and trimethylsilylacetylene **2l**, the reaction only afforded alka-1,4-diyn-4yl amine intermediate (*S*,*S*)-**5cl** (the diastereomer of (*R*,*S*)-**5ba** formed from **1b** and **2a**) and the corresponding allene (R_a)-**4cl** was not formed (Figure 3B-a). Even when (*S*,*S*)-**5cl** was treated under the standard conditions, allene (R_a)-**4cl** was not formed. Furthermore, the reaction of 3-TBS-2-alkynal **1a** with trimethylsilylacetylene **2l** also afforded the corresponding amine (*R*,*S*)-**5al** and allene (R_a)-**4al** was not formed, which is in agreement with the fact that when (*R*,*S*)-**5al** was treated under the standard conditions, allene (R_a)-**4al** was not formed (Figure 3B-b). The same results were observed for the reaction of 3-TMS-2-alkynal **1b** with TBS-acetylene **2m** (Figure 3B-c). Thus, it seems that the silyl-substituted terminal alkynes cannot be converted to the corresponding allenes. To further investigate the role of the silyl group in the starting alkynal, a sterically smaller phenethyl group was applied instead of the bulky trialkylsilyl moiety in the aldehyde (Figure 3B-d). Interestingly, the allenyne product (R_a)-**4da** was obtained, albeit in 14 % yield with 98 % *ee* together with 17 % yield of the 1,4-diyn-3-yl amine intermediate (*S*,*S*)-**5da** remaining. What's more, the reaction of 2-alkynal **1c** and propargylic alcohol **2a**, in the presence of (*S*)-**3**, or **1e** and **2n** in the presence of (*R*)-**3**, which both have the same substituent on the C_{sp} atom afforded the allene (R_a)-**4ca** or (S_a)-**4en** with an excellent *ee* (Figure 3B-(e,f)). The high *ee* was maintained when 1,4-diyn-3-yl amine intermediates (*S*,*S*)-**5da** and (*S*)-**5ca** as well as (*R*)-**5en** were treated under the standard conditions. All these results suggested that (1) 2-alkynals are NOT limited to the 3-silyl substituted ones, (2) the high regioselectivity of this EATA reaction is not dominated by the steric effect of the 3-substituent in the conjugated 2-alkynals, and (3) only one triple bond in (*S*)-**5ca** as well as (*R*)-**5en** underwent the 1,5-H transfer and β-*N*-elimination to form the allene unit.

Based on these results and previous reports,[6d,8c] a mechanism involving CuCl/$CuBr_2$ co-catalyzed system is proposed

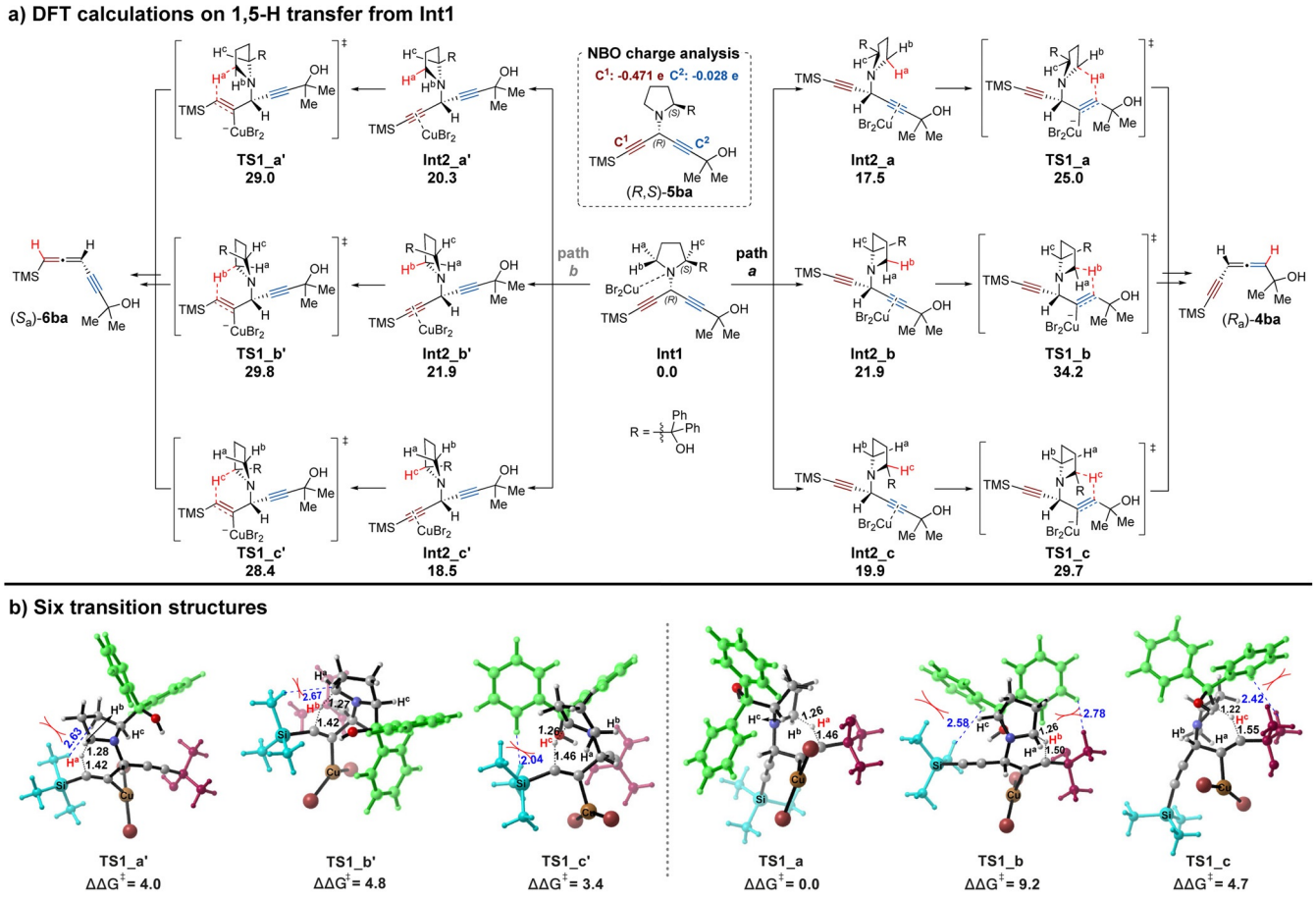


Figure 4. The 1,5-H transfer step from **Int1**. The energies in (a) (Δ*G*) are in kcal mol^{-1} with respect to **Int1**; the energies in (b) (ΔΔ*G*) are in kcal mol^{-1} with respect to **TS1_a**. Bond lengths are given in Ångstroms.

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as shown in Figure 3C. Initially, 2-alkynal **1** reacts with proline (*S*)-**3** to afford the reactive iminium ion **I**, which would be trapped by the in situ generated alkynylmetal species **II** to produce highly enantioenriched 1,4-diyn-3-yl amine (*R*,*S*)-**5**. Then the intermediate (*R*,*S*)-**5** undergoes sequential regio- and stereoselective 1,5-H transfer and *trans* β-*N*-elimination to yield the chiral allenyne (R_a)-**4** in the presence of $CuBr_2$.

In order to further rationalize the observed unique regio- and enantio-selectivity, DFT calculations were carried out to explore the reaction pathway for the transformation of propargylic amine (*R*,*S*)-**5ba** to allenyne (R_a)-**4ba** (Figure 3A) (see Supporting Information for details of computational methods). The reported energies are Gibbs energies that incorporate the effect of 1,4-dioxane solution ($\varepsilon = 2.21$).

The calculation starts from (*R*,*S*)-**5ba**-$CuBr_2$ complex **Int1** (selected as the free energy reference), in which $CuBr_2$ coordinates to the nitrogen atom. As shown in Figure 4a, both C–C triple bonds of (*R*,*S*)-**5ba** could be activated by $CuBr_2$ to trigger the subsequent 1,5-H transfer.[17] Notably, there are three different hydrogens (H^a, H^b, H^c in **Int1** of Figure 4a) adjacent to the *N* atom in the (*S*)-diphenylprolinol moiety that is in possible positions for the 1,5-H transfer. Thus, six transition structures are located for the competing regioselective 1,5-H transfer, which are denoted as **TS1_a–c** and **TS1_a′–c′**, separately (Figure 4a). The isomerization of **Int1** to **Int2_a–c** (path *a*, Figure 4a), in which $CuBr_2$ coordinates with the hydroxyalkyl-substituted triple bond, is endergonic by at least 17.5 kcal mol^{-1}, providing the precursors (**Int2_a–c**) for the 1,5-H transfer to the hydroxyalkyl-substituted triple bond. On the other hand, TMS- substituted triple bond could also be activated by $CuBr_2$, leading to the formation of the 1,5-H transfer precursors **Int2_a′–c′** (path *b*, Figure 4a). The subsequent 1,5-H transfer is realized via six-membered cyclic transition structure **TS1_a–c** and **TS1_a′–c′**. Among the six transition structures, **TS1_a**, which is referring to the 1,5-H transfer of the methylene hydrogen (H^a) to the hydroxyalkyl-substituted triple bond, is found to be the most favorable one, due to the lowest free energy barrier of 25.0 kcal mol^{-1}. The results could interpret the formation of the highly regioselective of product allenyne (R_a)-**4ba** via the 1,5-H transfer to the hydroxyalkyl-substituted triple bond in the experiment (No regioisomer (S_a)-**6ba** was observed, Figure 3A). Moreover, the fact that the methylene hydrogen in (*S*)-diphenylprolinol moiety participates in the 1,5-H transfer is in agreement with the deuterium-labeling experiment in our previous work.[8c]

To investigate the origin of the regioselectivity, natural bond orbital (NBO) charge analysis has been performed for (*R*,*S*)-**5ba**. As shown in Figure 4a, due to the influence of the different substituents (trimethylsilyl and hydroxylalkyl), the natural atomic charge on C^1 is much more negative than that on C^2 (−0.471 e vs. −0.028 e), indicating that 1,5-hydride migration to C^2 is more favorable.

Careful examination of the geometries of six competing 1,5-H transfer transition structures (Figure 4b) reveals that the steric effect caused by (*S*)-diphenylprolinol moiety is the key factor governing the selectivity. For **TS1_a**, the (*S*)-diphenylprolinol moiety is placed at the less hindered position. However, in the other five transition structures except **TS1_a**, there exist obvious steric repulsions between phenyl group or pyrrolidine unit in (*S*)-diphenylprolinol

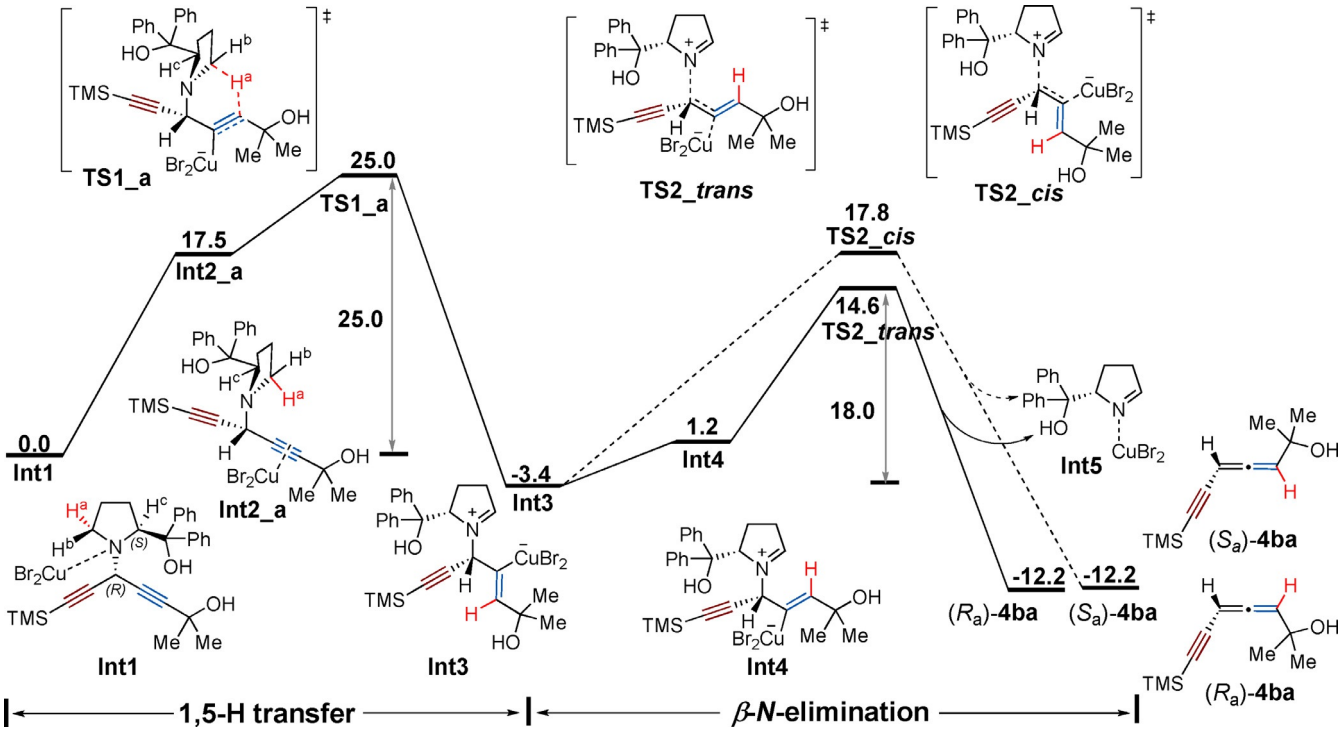


Figure 5. Free energy profile (Δ*G*, in kcal mol^{-1}) for the favorable 1,5-H transfer and subsequent β-*N*-elimination from **Int1**. All energies are in kcal mol^{-1} with respect to **Int1**. Bond lengths are given in Ångstroms.

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moiety and TMS or hydroxyalkyl substituent of the triple bonds. For **TS1_b**, both TMS and hydroxyalkyl substituents suffer steric hindrances with both phenyl groups in the (*S*)-diphenylprolinol moiety (The nearest H···H distances are 2.58 Å and 2.78 Å, separately), thus, contributing to the least stability of **TS1_b** (9.2 kcal mol^{-1} higher in energy than **TS1_a**, Figure 4b). Furthermore, the steric hindrance between phenyl group and hydroxyalkyl substituent (The nearest H···H distance is 2.42 Å) in **TS1_c** and the steric hindrance between phenyl group and TMS substituent (The nearest H···H distance is 2.04 Å) in **TS1_c′** destabilize both structures, making them less favorable by 4.7 and 3.4 kcal mol^{-1} referring to **TS1_a** (Figure 4b). While for **TS1_a′** and **TS1_b′**, the stability of both structures is influenced by the steric interaction between pyrrolidine unit and TMS substituent (The nearest H···H distance are 2.63 Å and 2.67 Å, respectively), which could account for the less stability of **TS1_a′** and **TS1_b′** by 4.0 and 4.8 kcal mol^{-1} referring to **TS1_a** (Figure 4b). Therefore, with no obvious steric hindrance, **TS1_a** becomes the most favorable transition structure.

The computed free energy profile is shown in Figure 5. The most favorable 1,5-H transfer led to the formation of *N*-allylic iminium-$CuBr_2$ intermediate **Int3**, which is lower in energy by 3.4 kcal mol^{-1} with respect to **Int1**. (R_a)-**4ba** was obtained through *trans* β-*N*-elimination, while *cis* β-*N*-elimination led to the formation of (S_a)-**4ba**. The transition states for both *trans* and *cis* elimination have been located as **TS2_*trans*** and **TS2_*cis***. The calculations show that **TS2_*trans*** is favored over **TS2_*cis*** by 3.2 kcal mol^{-1} (14.6 kcal mol^{-1} vs. 17.8 kcal mol^{-1}), indicating that β-*N*-elimination step overwhelmingly proceeds via *trans* manner to provide (R_a)-**4ba**, which is consistent with the experimental observation of (R_a)-**4ba** with > 99 % *ee* as the only final product.

The transformation of (*S*,*S*)-**5cl** (Figure 3B-a) was also investigated by DFT calculations, the free energy profile is shown in Figure 6 (The two most favorable transition

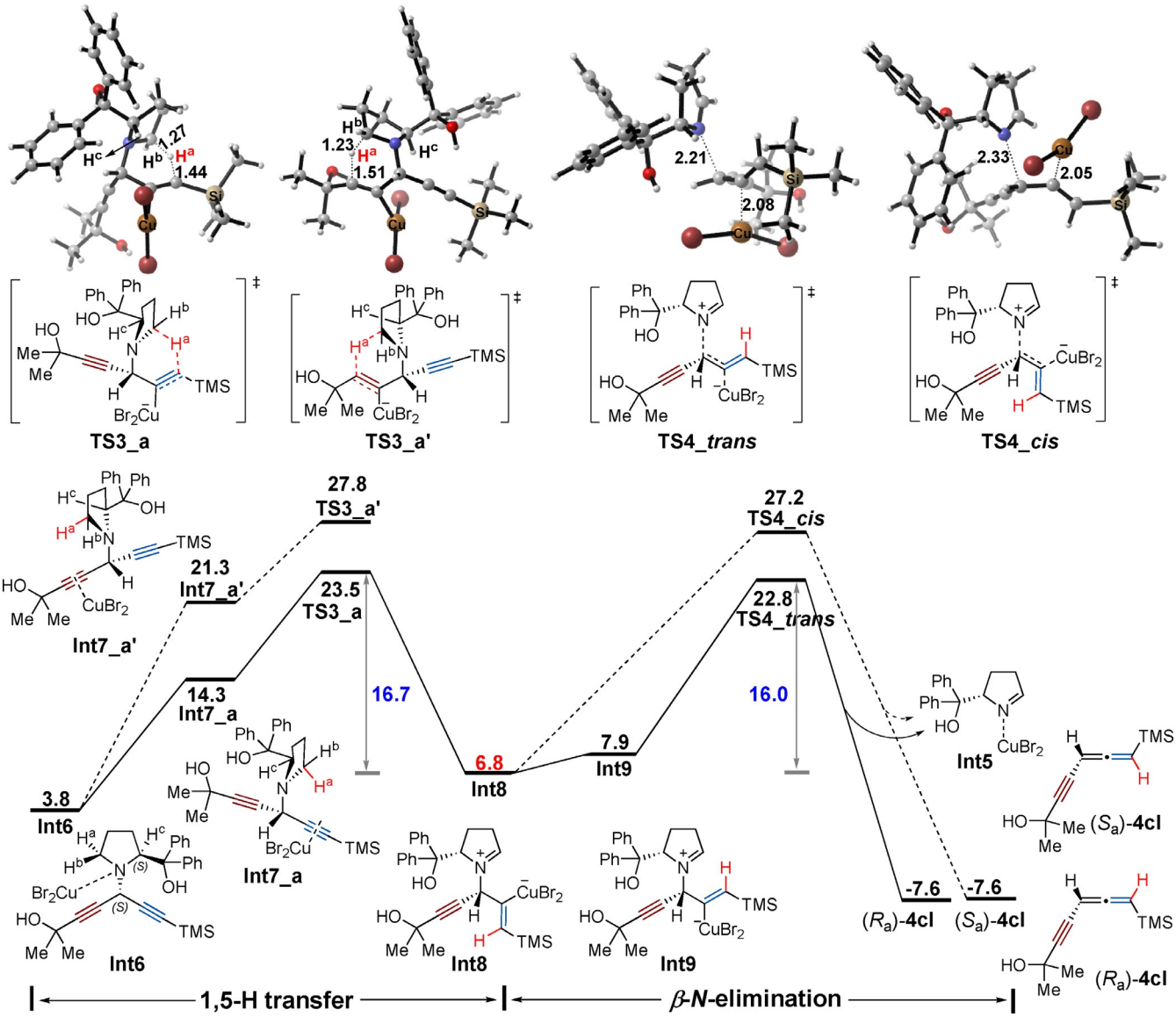


Figure 6. Free energy profile (Δ*G*, in kcal mol^{-1}) for the favorable 1,5-H transfer and subsequent β-*N*-elimination from **Int6**. All energies are in kcal mol^{-1} with respect to **Int1**. Bond lengths are given in Ångstroms.

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structures for the 1,5-H transfer to both C–C triple bond are depicted in Figure 6, and see Figure S4 in Supporting Information for all six competing 1,5-H transfer transition structures). Different from the 1,5-H transfer of (*R*,*S*)-**5ba**-$CuBr_2$ (Figure 3A and Figure 5), the 1,5-H transfer step from (*S*,*S*)-**5cl**-$CuBr_2$ complex **Int6** is endothermic, leading to the formation of the less stable intermediate **Int8**, which is higher in free energy of 3.0 kcal mol^{-1} than **Int6**. Thus, the 1,5-H transfer in this process is reversible. From iminium intermediate **Int8**, *trans* β-*N*-elimination via **TS4_*trans*** to produce allenyne product (R_a)-**4cl** requires a free energy barrier of 16.0 kcal mol^{-1}, which is slightly lower than that of the reverse 1,5-H transfer via **TS3_a** (16.7 kcal mol^{-1}), indicating the possibility of both processes. These results could account for the experimental observation of 33% recovery of (*S*,*S*)-**5cl**. However, it should be noted that the silyl-substituted allenynes [(R_a) or (S_a)-**4cl**, **4al**, **4bm**] are not observed experimentally in all cases (Figure 3B-(a–c)), most probably due to the decomposition or transformation to other unknown species.

Conclusion

In conclusion, we have developed a highly efficient CuCl/$CuBr_2$ co-catalyzed enantioselective allenation (EATA) reaction of 2-alkynals by using the commercially available (*S*)- or (*R*)-α,α-diphenylprolinol involving the highly regio- and stereo-selective 1,5-H transfer of 1,4-diyn-3-yl amine intermediates. This method could tolerate a wide range of functional groups including the chiral secondary free propargylic alcohols, and provides a series of allenynes with excellent enantioselectivities (up to >99% *ee*). The enantioenriched allenynes can be readily transformed into various building blocks. This protocol has been successfully applied to the first highly enantioselective total synthesis of linear allenic natural product (−)-scorodonin and its enantiomer. Further mechanistic studies involving both experiments and DFT calculations revealed that the configuration of proline is responsible for the stereo- and regioselectivity observed in this transformation.

Acknowledgements

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: alkynes · allenes · copper · enantioselectivity · regioselectivity

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